

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(-\tilde{\mu} y_d^{i3i4} y_e^{pr} y_u^{i1i2} \overline{a_e^{pr}} LF_{3,1,0} [m_l^P, m_e^r] + 3 \tilde{\mu} y_d^{i3i4} y_e^{pr} y_u^{i1i2} \overline{a_e^{pr}} LF_{4,1,-1} [m_l^P, m_e^r] - \right. \\
& 2 \tilde{\mu} y_d^{i3i4} y_e^{pr} y_u^{i1i2} \overline{a_e^{pr}} LF_{5,1,-2} [m_l^P, m_e^r] - 3 \tilde{\mu} y_d^{pr} y_d^{i3i4} y_u^{i1i2} \overline{a_d^{pr}} LF_{3,1,0} [m_q^P, m_d^r] + \\
& 9 \tilde{\mu} y_d^{pr} y_d^{i3i4} y_u^{i1i2} \overline{a_d^{pr}} LF_{4,1,-1} [m_q^P, m_d^r] - 6 \tilde{\mu} y_d^{pr} y_d^{i3i4} y_u^{i1i2} \overline{a_d^{pr}} LF_{5,1,-2} [m_q^P, m_d^r] - \\
& 3 \tilde{\mu} y_d^{i3i4} y_u^{pr} y_u^{i1i2} \overline{a_u^{pr}} LF_{3,1,0} [m_u^r, m_q^P] + 9 \tilde{\mu} y_d^{i3i4} y_u^{pr} y_u^{i1i2} \overline{a_u^{pr}} LF_{4,1,-1} [m_u^r, m_q^P] - \\
& 6 \tilde{\mu} y_d^{i3i4} y_u^{pr} y_u^{i1i2} \overline{a_u^{pr}} LF_{5,1,-2} [m_u^r, m_q^P] - m_1 \tilde{\mu} g_1^2 y_d^{i3i4} y_u^{i1i2} LF_{3,1,0} [\tilde{\mu}, m_1] + \\
& 3 m_1 \tilde{\mu} g_1^2 y_d^{i3i4} y_u^{i1i2} LF_{4,1,-1} [\tilde{\mu}, m_1] - 2 m_1 \tilde{\mu} g_1^2 y_d^{i3i4} y_u^{i1i2} LF_{5,1,-2} [\tilde{\mu}, m_1] - \\
& 3 m_2 \tilde{\mu} g_2^2 y_d^{i3i4} y_u^{i1i2} LF_{3,1,0} [\tilde{\mu}, m_2] + 9 m_2 \tilde{\mu} g_2^2 y_d^{i3i4} y_u^{i1i2} LF_{4,1,-1} [\tilde{\mu}, m_2] - \\
& 6 m_2 \tilde{\mu} g_2^2 y_d^{i3i4} y_u^{i1i2} LF_{5,1,-2} [\tilde{\mu}, m_2] - \frac{1}{36} m_1 \tilde{\mu} g_1^2 y_d^{i3i4} y_u^{i1i2} LF_{2,2,1,-1} [m_1, m_d^{i4}, m_q^{i3}] + \\
& \frac{1}{18} m_1 \tilde{\mu} g_1^2 y_d^{i3i4} y_u^{i1i2} LF_{2,2,1,-1} [m_1, m_q^{i1}, m_u^{i2}] - \frac{1}{36} m_1 \tilde{\mu} g_1^2 y_d^{i3i4} y_u^{i1i2} \\
& LF_{2,2,1,-1} [m_1, m_q^{i3}, m_d^{i4}] + \frac{1}{18} m_1 \tilde{\mu} g_1^2 y_d^{i3i4} y_u^{i1i2} LF_{2,2,1,-1} [m_1, m_u^{i2}, m_q^{i1}] + \\
& \frac{1}{6} m_1 \tilde{\mu} g_1^2 y_d^{i3i4} y_u^{i1i2} LF_{2,2,1,-1} [m_1, \tilde{\mu}, m_d^{i4}] - \frac{1}{12} m_1 \tilde{\mu} g_1^2 y_d^{i3i4} y_u^{i1i2} LF_{2,2,1,-1} [m_1, \tilde{\mu}, m_q^{i1}] + \\
& \frac{1}{12} m_1 \tilde{\mu} g_1^2 y_d^{i3i4} y_u^{i1i2} LF_{2,2,1,-1} [m_1, \tilde{\mu}, m_q^{i3}] + \frac{1}{3} m_1 \tilde{\mu} g_1^2 y_d^{i3i4} y_u^{i1i2} LF_{2,2,1,-1} [m_1, \tilde{\mu}, m_u^{i2}] + \\
& \frac{3}{4} m_2 \tilde{\mu} g_2^2 y_d^{i3i4} y_u^{i1i2} LF_{2,2,1,-1} [m_2, \tilde{\mu}, m_q^{i1}] + \frac{3}{4} m_2 \tilde{\mu} g_2^2 y_d^{i3i4} y_u^{i1i2} LF_{2,2,1,-1} [m_2, \tilde{\mu}, m_q^{i3}] + \\
& \frac{2}{3} m_3 \tilde{\mu} g_3^2 y_d^{i3i4} y_u^{i1i2} LF_{2,2,1,-1} [m_3, m_d^{i4}, m_q^{i3}] + \frac{2}{3} m_3 \tilde{\mu} g_3^2 y_d^{i3i4} y_u^{i1i2} \\
& LF_{2,2,1,-1} [m_3, m_q^{i1}, m_u^{i2}] + \frac{2}{3} m_3 \tilde{\mu} g_3^2 y_d^{i3i4} y_u^{i1i2} LF_{2,2,1,-1} [m_3, m_q^{i3}, m_d^{i4}] + \\
& \frac{2}{3} m_3 \tilde{\mu} g_3^2 y_d^{i3i4} y_u^{i1i2} LF_{2,2,1,-1} [m_3, m_u^{i2}, m_q^{i1}] - \frac{1}{2} \tilde{\mu} y_d^{i1r} y_d^{i3i4} y_u^{pi2} \overline{a_d^{pr}} \\
& LF_{2,1,1,0} [m_d^r, m_q^P, \tilde{\mu}] + \frac{1}{2} \tilde{\mu} y_d^{i1r} y_d^{i3i4} y_u^{pi2} \overline{a_d^{pr}} LF_{3,1,1,-1} [m_d^r, m_q^P, \tilde{\mu}] + \\
& \frac{1}{4} \tilde{\mu} y_d^{i1r} y_d^{i3i4} y_u^{pi2} \overline{a_d^{pr}} LF_{2,2,1,-1} [m_d^r, \tilde{\mu}, m_q^P] + \frac{1}{18} m_1 \tilde{\mu} g_1^2 y_d^{i3i4} y_u^{i1i2} \\
& LF_{2,1,1,0} [m_d^{i4}, m_1, m_q^{i3}] - \frac{1}{18} m_1 \tilde{\mu} g_1^2 y_d^{i3i4} y_u^{i1i2} LF_{3,1,1,-1} [m_d^{i4}, m_1, m_q^{i3}] - \\
& \frac{4}{3} m_3 \tilde{\mu} g_3^2 y_d^{i3i4} y_u^{i1i2} LF_{2,1,1,0} [m_d^{i4}, m_3, m_q^{i3}] + \frac{4}{3} m_3 \tilde{\mu} g_3^2 y_d^{i3i4} y_u^{i1i2} \\
& LF_{3,1,1,-1} [m_d^{i4}, m_3, m_q^{i3}] - \frac{1}{2} \tilde{\mu} y_d^{i1r} y_d^{i3i4} y_u^{pi2} \overline{a_d^{pr}} LF_{2,1,1,0} [m_q^P, m_d^r, \tilde{\mu}] + \\
& \frac{1}{2} \tilde{\mu} y_d^{i1r} y_d^{i3i4} y_u^{pi2} \overline{a_d^{pr}} LF_{3,1,1,-1} [m_q^P, m_d^r, \tilde{\mu}] - \frac{1}{2} \tilde{\mu} y_d^{pi4} y_u^{i1i2} y_u^{i3r} \overline{a_u^{pr}} \\
& LF_{2,1,1,0} [m_q^P, m_u^r, \tilde{\mu}] + \frac{1}{2} \tilde{\mu} y_d^{pi4} y_u^{i1i2} y_u^{i3r} \overline{a_u^{pr}} LF_{3,1,1,-1} [m_q^P, m_u^r, \tilde{\mu}] + \\
& \frac{1}{4} \tilde{\mu} y_d^{i1r} y_d^{i3i4} y_u^{pi2} \overline{a_d^{pr}} LF_{2,2,1,-1} [m_q^P, \tilde{\mu}, m_d^r] + \frac{1}{4} \tilde{\mu} y_d^{pi4} y_u^{i1i2} y_u^{i3r} \overline{a_u^{pr}} \\
& LF_{2,2,1,-1} [m_q^P, \tilde{\mu}, m_u^r] - \frac{1}{9} m_1 \tilde{\mu} g_1^2 y_d^{i3i4} y_u^{i1i2} LF_{2,1,1,0} [m_q^{i1}, m_1, m_u^{i2}] + \\
& \frac{1}{9} m_1 \tilde{\mu} g_1^2 y_d^{i3i4} y_u^{i1i2} LF_{3,1,1,-1} [m_q^{i1}, m_1, m_u^{i2}] - \frac{4}{3} m_3 \tilde{\mu} g_3^2 y_d^{i3i4} y_u^{i1i2} \\
& LF_{2,1,1,0} [m_q^{i1}, m_3, m_u^{i2}] + \frac{4}{3} m_3 \tilde{\mu} g_3^2 y_d^{i3i4} y_u^{i1i2} LF_{3,1,1,-1} [m_q^{i1}, m_3, m_u^{i2}] + \\
& \frac{1}{18} m_1 \tilde{\mu} g_1^2 y_d^{i3i4} y_u^{i1i2} LF_{2,1,1,0} [m_q^{i3}, m_1, m_d^{i4}] - \frac{1}{18} m_1 \tilde{\mu} g_1^2 y_d^{i3i4} y_u^{i1i2} \\
& LF_{3,1,1,-1} [m_q^{i3}, m_1, m_d^{i4}] - \frac{4}{3} m_3 \tilde{\mu} g_3^2 y_d^{i3i4} y_u^{i1i2} LF_{2,1,1,0} [m_q^{i3}, m_3, m_d^{i4}] + \\
& \frac{4}{3} m_3 \tilde{\mu} g_3^2 y_d^{i3i4} y_u^{i1i2} LF_{3,1,1,-1} [m_q^{i3}, m_3, m_d^{i4}] - \frac{1}{2} \tilde{\mu} y_d^{pi4} y_u^{i1i2} y_u^{i3r} \overline{a_u^{pr}} \\
& LF_{2,1,1,0} [m_u^r, m_q^P, \tilde{\mu}] + \frac{1}{2} \tilde{\mu} y_d^{pi4} y_u^{i1i2} y_u^{i3r} \overline{a_u^{pr}} LF_{3,1,1,-1} [m_u^r, m_q^P, \tilde{\mu}] + \\
& \frac{1}{4} \tilde{\mu} y_d^{pi4} y_u^{i1i2} y_u^{i3r} \overline{a_u^{pr}} LF_{2,2,1,-1} [m_u^r, \tilde{\mu}, m_q^P] - \frac{1}{9} m_1 \tilde{\mu} g_1^2 y_d^{i3i4} y_u^{i1i2} \\
& LF_{2,1,1,0} [m_u^{i2}, m_1, m_q^{i1}] + \frac{1}{9} m_1 \tilde{\mu} g_1^2 y_d^{i3i4} y_u^{i1i2} LF_{3,1,1,-1} [m_u^{i2}, m_1, m_q^{i1}] - \\
& \frac{4}{3} m_3 \tilde{\mu} g_3^2 y_d^{i3i4} y_u^{i1i2} LF_{2,1,1,0} [m_u^{i2}, m_3, m_q^{i1}] + \frac{4}{3} m_3 \tilde{\mu} g_3^2 y_d^{i3i4} y_u^{i1i2} \\
& LF_{3,1,1,-1} [m_u^{i2}, m_3, m_q^{i1}] + \frac{1}{27} m_1 \tilde{\mu} g_1^2 y_d^{i1i4} y_u^{i3i2} LF_{1,1,1,1,0} [m_1, m_d^{i4}, m_q^{i3}, \tilde{\mu}] + \\
& \frac{4}{27} m_1 \tilde{\mu} g_1^2 (-3 y_d^{i3i4} y_u^{i1i2} + y_d^{i1i4} y_u^{i3i2}) LF_{1,1,1,1,0} [m_1, m_d^{i4}, m_u^{i2}, \tilde{\mu}] + \\
& \frac{1}{27} m_1 \tilde{\mu} g_1^2 y_d^{i1i4} y_u^{i3i2} LF_{1,1,1,1,0} [m_1, m_q^{i1}, m_q^{i3}, \tilde{\mu}] - \\
& \frac{2}{27} m_1 \tilde{\mu} g_1^2 y_d^{i1i4} y_u^{i3i2} LF_{1,1,1,1,0} [m_1, m_q^{i1}, m_u^{i2}, \tilde{\mu}] - \\
& m_2 \tilde{\mu} g_2^2 y_d^{i3i4} y_u^{i1i2} LF_{1,1,1,1,0} [m_2, m_q^{i1}, m_q^{i3}, \tilde{\mu}] - \\
& \frac{8}{9} m_3 \tilde{\mu} g_3^2 y_d^{i1i4} y_u^{i3i2} LF_{1,1,1,1,0} [m_3, m_d^{i4}, m_q^{i3}, \tilde{\mu}] - \\
& \frac{8}{9} m_3 \tilde{\mu} g_3^2 y_d^{i1i4} y_u^{i3i2} LF_{1,1,1,1,0} [m_3, m_d^{i4}, m_u^{i2}, \tilde{\mu}] - \\
& \frac{8}{9} m_3 \tilde{\mu} g_3^2 y_d^{i1i4} y_u^{i3i2} LF_{1,1,1,1,0} [m_3, m_q^{i1}, m_q^{i3}, \tilde{\mu}] - \\
& \left. \frac{8}{9} m_3 \tilde{\mu} g_3^2 y_d^{i1i4} y_u^{i3i2} LF_{1,1,1,1,0} [m_3, m_q^{i1}, m_u^{i2}, \tilde{\mu}] \right)
\end{aligned}$$